

Carbon Pricing in Reverse: Evidence from Virginia's Exit and Re-Entry into RGGI

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Abstract

[PENDING DATA: Abstract.] Virginia is the only U.S. state to enter, exit, and re-enter a carbon cap-and-trade program (the Regional Greenhouse Gas Initiative, RGGI) within a five-year span. Because the exit was driven by a change in gubernatorial administration rather than by economic conditions, the timing of treatment is plausibly exogenous. We use synthetic control and event-study difference-in-differences methods on public data to estimate the causal effect of Virginia's RGGI participation on retail electricity prices, wholesale prices, and CO₂ emissions, and we test whether the effect of exiting a carbon market is symmetric.

Keywords: carbon pricing; cap-and-trade; RGGI; synthetic control; electricity markets; emissions leakage; policy reversal.

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1 Introduction

[TODO: Phase 5 prose.] Carbon cap-and-trade is the dominant market-based climate policy instrument in the United States, yet credible causal evidence on its effects is scarce because clean counterfactuals are rare. Virginia provides a unique natural experiment.

Research questions. **[TODO: prose around the four questions:]** (i) the causal effect of Virginia’s RGGI exit on retail prices, wholesale prices, and CO₂emissions; (ii) *symmetry of exit vs. entry effects*; (iii) *PJM emissions leakage*; (iv) *whether re-entry replicates the entry effects*.

Contribution. **[TODO: prose:]** first empirical study of a carbon-market *exit*; a test for asymmetry/hysteresis exploiting a rare reversal; within-RTO (PJM) leakage estimates under sub-national pricing; a paired entry/exit/re-entry design on a single treated unit. Early illustrative evidence appears in [Murray and Maniloff \(2015\)](#).

2 Institutional Background

2.1 RGGI Overview

[TODO: Phase 5 prose:] 2009 launch, program reviews, current membership; quarterly allowance auctions, compliance obligations, Cost Containment Reserve (CCR), Emissions Containment Reserve (ECR). Third Program Review changes effective 2027 (tighter cap trajectory, higher minimum reserve price, elimination of offsets).

2.2 Virginia’s RGGI Timeline

Table 1: Virginia’s RGGI participation timeline.

Date	Event	Political context
Apr. 2020	Clean Economy Act & Community Flood Preparedness Act signed	Democratic trifecta
Jan. 2021	Virginia begins RGGI participation	First compliance period
Jan. 2022	Youngkin inaugurated; signals intent to withdraw	Republican governor, split legislature
Jun. 2023	Air Pollution Control Board votes to repeal RGGI regulation	Executive-driven withdrawal
Dec. 2023	Last Virginia participation in RGGI auction	Formal exit
Jan. 2024–Jun. 2026	Virginia outside RGGI	Non-participation period
2025	Spanberger (D) elected; legislation to rejoin signed	Democratic trifecta restored
Jul. 2026	Virginia re-enters RGGI	Second participation period

Sources: **[TODO: Phase 5 – cite Clean Economy Act; APCB repeal; RGGI auction records.]**

2.3 The Dominion Energy Context

[TODO: Phase 5 prose, public sources only:] Dominion-dominated regulated market; RGGI costs passed through via a rider (**[TODO: rider \$/month from public SCC docket]**); rider removed on exit, to be reinstated on re-entry. Concurrent factors: residential rate increases, data center load growth, IRA-driven clean-energy investment. *No Dominion internal data are used; rider figures are drawn from public SCC filings.*

3 Literature Review

[TODO: Phase 5 prose, organized in four strands:]

- **Empirical effects of carbon pricing:** Murray and Maniloff (2015); **[TODO: Fell & Maniloff 2018; Yan 2021; Song & Hochman 2025.]**
- **Pass-through & incidence:** **[TODO: Fabra & Reguant 2014; Hintermann 2016; Fowlie, Reguant & Ryan 2016; Bushnell et al. 2013.]**
- **Policy reversal & asymmetry:** **[TODO: Greenstone 2002; Coglianese et al. 2020; Walker 2013.]**
- **Methodology:** Abadie et al. (2010); **[TODO: Abadie et al. 2015; Abadie 2021;**

]Arkhangelsky et al. (2021)[**TODO: ; Cattaneo et al. 2021; Callaway & Sant’Anna 2021; Cameron et al. 2008.**]

4 Theoretical Framework

4.1 A Carbon Adder in a Regulated Market

We model the retail price of electricity as

$$p_r = c_g + c_T + \pi_{\text{allow}} \cdot e + \mu, \quad (1)$$

where c_g is generation cost (fuel, O&M), c_T is transmission and distribution cost, π_{allow} is the allowance price (\$/MWh once scaled by the emissions rate e in tons/MWh of the marginal generator), and μ is the regulated markup. Removing the carbon adder ($\pi_{\text{allow}} \rightarrow 0$) reduces p_r mechanically through the rider; the wholesale-price effect depends on whether RGGI-state generators are marginal in PJM dispatch.

4.2 Predictions

Prediction 1 (Retail prices). *Retail prices fall by approximately the rider amount on exit and rise by approximately the new rider on re-entry.*

Prediction 2 (Wholesale LMPs). *Ambiguous: Virginia generators offer lower prices, but if inframarginal the LMP may not move.*

Prediction 3 (Virginia emissions). *Rise if the adder was constraining dispatch of Virginia fossil generators.*

Prediction 4 (System-wide emissions). *Ambiguous: depends on whether Virginia generation displaces cleaner or dirtier out-of-state generation (leakage).*

Prediction 5 (Asymmetry/hysteresis). *Entry effects may exceed exit effects if 2021–2023 clean investments are irreversible, efficiency gains persist, or adjustments have ratchet properties.*

5 Data

[**TODO: Phase 5 prose.**] *Public sources only.*

Table 2: Data sources (public). **[TODO: Regenerated from config in Phase 2–3.]**

Variable	Source	Frequency	Coverage
Retail electricity prices	EIA Form 861/861M	Monthly/Annual	States, 2018–2026
Wholesale prices (LMPs)	PJM Data Miner 2.0	Hourly → monthly	PJM nodes/zones
CO ₂ emissions	EPA CEMS (CAMPD)	Hourly → monthly	U.S. fossil units
Generation by fuel	EIA 923 / EPA CEMS	Monthly	U.S. generators
Allowance auction prices	RGGI Inc. reports	Quarterly	2008–present
Natural gas prices	EIA (Henry Hub, Transco Z5)	Daily/Monthly	National/Regional
Weather (HDD/CDD)	NOAA ISD/GHCN	Daily	By state
Economic controls	BEA / Census	Annual/Quarterly	States
Dominion rider	Public SCC dockets	Per filing	Virginia

Notes: Public sources only; no Dominion internal data. **[TODO: Add access dates/vintages in Phase 2.]**

5.1 Sample Construction

[TODO: Phase 5 prose:] unit = state (PJM zone for wholesale); time = month (2018–2026); treated = Virginia. Control groups: non-RGGI PJM states (WV, OH, PA, KY, IN, NC, TN) and continuous-RGGI states (MD, DE, NJ, CT, MA, ME, NH, VT, RI, NY).

6 Empirical Strategy

6.1 Synthetic Control (Primary)

[TODO: Phase 5 prose:] construct synthetic Virginia matching pre-treatment outcomes and covariates, separately for the entry (pre: Jan. 2018–Dec. 2020) and exit (pre: Jan. 2021–Sep. 2023) windows. Inference via placebo/permutation tests and conformal prediction intervals.

6.2 Event-Study Difference-in-Differences

$$Y_{st} = \alpha_s + \gamma_t + \sum_{\tau \neq -1} \beta_\tau \mathbf{1}[s = \text{VA}] \mathbf{1}[t - t^* = \tau] + X'_{st} \delta + \varepsilon_{st}, \quad (2)$$

with state fixed effects α_s , month fixed effects γ_t , time-varying controls X_{st} (gas prices, HDD, CDD), and dynamic effects β_τ relative to $\tau = -1$. Standard errors clustered at the state level; wild cluster bootstrap given the small number of clusters. Estimated separately for the

exit ($t^* = \text{Dec. 2023/Jan. 2024}$), entry ($t^* = \text{Jan. 2021}$), and prospectively re-entry ($t^* = \text{Jul. 2026}$) events.

6.3 Synthetic Difference-in-Differences

[TODO: Phase 5 prose:] Arkhangelsky et al. (2021) as a bridge between SCM and DiD.

6.4 Extensions

[TODO: Phase 5 prose:] wholesale PJM-zone analysis (DOM zone treated); emissions leakage (VA-plant vs. PJM-wide emissions); formal asymmetry test $|\hat{\tau}^{\text{exit}}| = |\hat{\tau}^{\text{entry}}|$; distributional analysis if public microdata permit.

7 Threats to Identification and Robustness

Table 3: Threats and robustness checks.

Threat	Test / mitigation
Anticipation (exit signaled early 2022)	Alternative treatment dates (Jan. 2022, Jun. 2023, Dec. 2023); pre-trend tests.
Concurrent policy shocks (IRA, RPS)	Control for renewable capacity additions and federal subsidy proxies.
Gas price volatility	Control for regional gas hub prices; gas-adjusted outcomes.
Data center load growth	Control for large-customer load; use per-MWh prices.
SUTVA violation (PJM spillovers)	Estimate PJM-wide wholesale effects; bound spillovers.
Small- N inference	Permutation inference (SCM); wild cluster bootstrap (DiD).
Rider mechanicality	Decompose retail effect into rider (mechanical) and non-rider components.
Fleet composition changes	Control for capacity additions/retirements; shares vs. levels.

8 Results

[PENDING DATA: All results below are generated by scripts/R/ from public data and pulled in via \input/\includegraphics. No numbers are hand-entered.]

8.1 Exit Event (Dec. 2023)

[PENDING DATA: SCM gap plot, event-study coefficients, placebo distribution.]

8.2 Entry Event (Jan. 2021)

[PENDING DATA: SCM and event-study estimates for entry.]

8.3 Asymmetry Test

[PENDING DATA: Formal test of $|\hat{\tau}^{\text{exit}}| = |\hat{\tau}^{\text{entry}}|$.]

8.4 Emissions Leakage

[PENDING DATA: VA-plant vs. PJM-wide emissions; implied leakage rate.]

9 Expected Results and Interpretation

[TODO: Phase 5 prose around three pre-registered scenarios:]

Scenario A — Symmetric effects. Carbon-pricing effects are reversible; policy uncertainty may undermine long-run investment incentives.

Scenario B — Asymmetric (entry > exit). Irreversible clean investments create a ratchet; even temporary participation yields lasting benefits.

Scenario C — Leakage dominates. Sub-national design reshuffles generation across state lines without net reductions.

Re-entry (prospective). [PENDING DATA: The Jul. 2026 re-entry is in the future as of this draft. We pre-register the design (treatment date, donor pool, estimands, inference) and will report estimates once post-treatment data are available.]

10 Conclusion

[TODO: Phase 5 prose.]

References

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A Data Sources and Public Retrieval Methods

[TODO: Phase 3 – map each variable to its public source and access method.]

B Pre-Analysis Plan: Re-Entry (Jul. 2026)

[TODO: Phase 4 – pre-registered estimands, treatment date, donor pool, and inference for the re-entry event, fixed in advance of observing outcomes.]